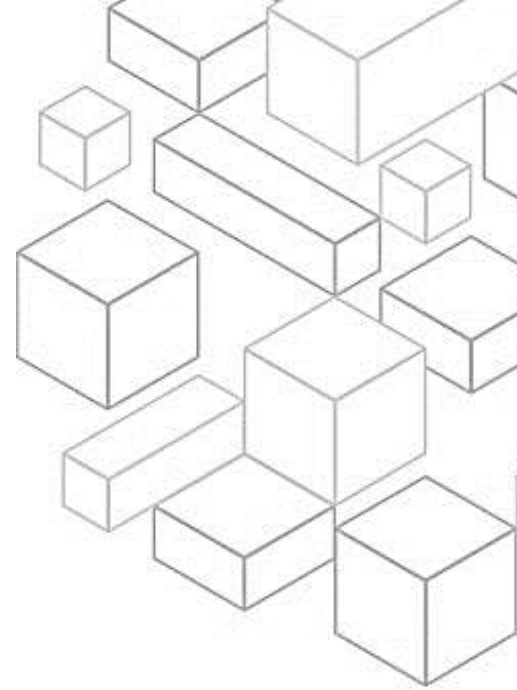




Module 8 AWS Well-Architected Cost Optimization

1.1 AWS Well-Architected

Welcome to module eight of AWS Well-Architected: Deep Dive on the Cost Optimization Pillar.

1.2 Learning objectives

In this module, you will get an overview of the cost optimization pillar of the AWS Well-Architected Framework. You will also learn the design principles and best practices of the cost optimization pillar.

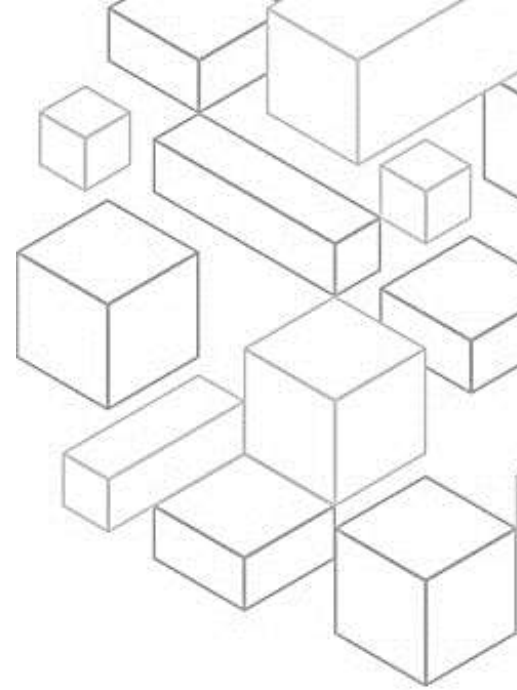
1.3 Cost Optimization

To begin, you will get an overview of the cost optimization pillar.

1.4 Pillars of Well-Architected

There are currently six pillars in the Well-Architected Framework: operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability. These pillars are the foundations of the architecture for your technology solutions in the cloud.

This module will focus on the cost optimization pillar.



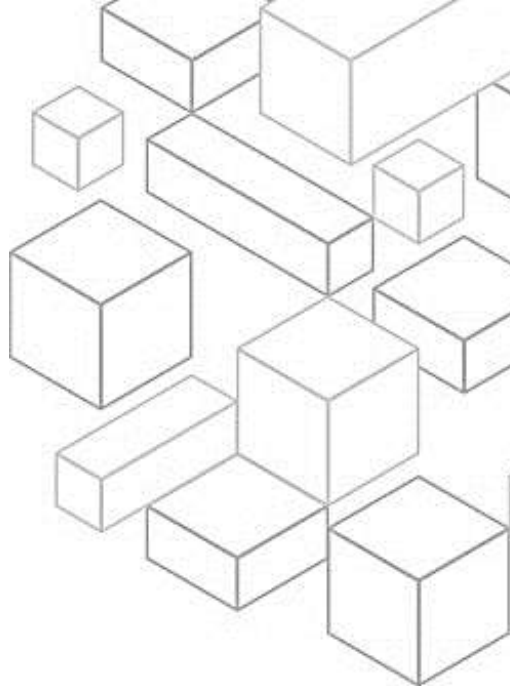
1.5 What is the cost optimization pillar?

What is the cost optimization pillar? Similar to the other pillars in the Well-Architected Framework, there are trade-offs to consider for cost optimization. One example is whether to optimize for speed to market or for cost. In some cases, it works best to optimize for speed-going to market quickly, shipping new features, or meeting a deadline-rather than investing in upfront cost optimization.

Design decisions are sometimes directed by haste rather than data. The temptation always exists to overcompensate, rather than spend time benchmarking for the most cost-optimal deployment. Overcompensation can lead to over-provisioned and under-optimized deployments. However, it might be a reasonable choice if you must lift and shift resources from your on-premises environment to the cloud and then optimize afterwards. Investing the right amount of effort in a cost optimization strategy up front can help you realize the economic benefits of the cloud more readily. You can adopt a consistent adherence to best practices and avoid unnecessary over-provisioning. The following sections provide techniques and best practices for the initial and ongoing implementation of Cloud Financial Management (CFM) and cost optimization for your workloads.

1.6 Cost Optimization Design Principles

Now that you understand what the cost optimization pillar is, you will dive deeper into the design principles of the cost optimization pillar.

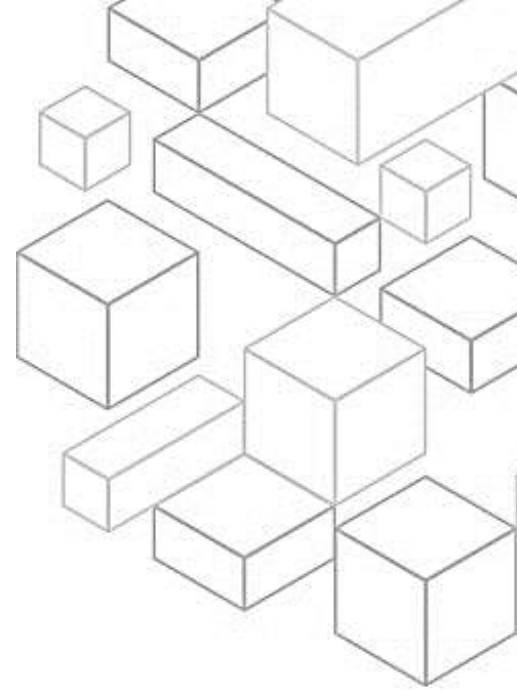


1.7 Cost optimization design principles

Several principles can help you with cost optimization in the cloud. First, to help achieve financial success and accelerate business value realization in the cloud, practice CFM. Your organization will need to dedicate the necessary time and resources for building capability in this new domain of technology and usage management. Similar to your security or operations capability, grow capability through knowledge building, programs, resources, and processes to help you become a cost-efficient organization.

Next, adopt a consumption model. Pay only for the computing resources you consume, and increase or decrease usage depending on business requirements. For example, development and test environments are typically only used for 8 hours a day during the workweek. You can stop these resources when they are not in use for a potential cost savings of 75 percent, or 40 hours compared to 168 hours. You can also measure overall efficiency. Monitor the business output of the workload and the costs associated with delivery. Use this data to understand the gains you make from increasing output, increasing functionality, and reducing cost.

Stop spending money on undifferentiated heavy lifting. AWS does the heavy lifting of data-center operations like racking, stacking, and powering servers. It also removes the operational burden of managing operating systems and applications with managed services. This helps you focus on your customers and business projects rather than on IT infrastructure. Finally, analyze and attribute expenditure. The cloud helps you accurately identify cost and usage of workloads to transparently attribute IT costs to revenue streams and workload owners. This helps measure return on investment (ROI) and gives workload owners an opportunity to optimize their resources and reduce costs.



1.8 Cost Optimization

Now that you understand the cost optimization design principles, you will learn about the cost optimization best practices.

1.9 Cost optimization best practice areas

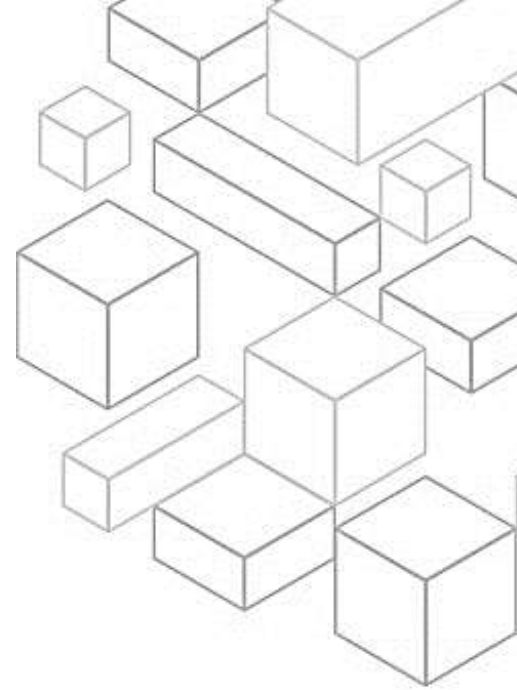
The cost optimization pillar is grouped into five areas of best practices. These include CFM, expenditure and usage awareness, cost-effective resources, management of demand and supply resources, and optimization over time.

1.10 Practice Cloud Financial Management

Practice cloud financial management is the first cost optimization best practice area.

1.11 Practice cloud financial management

Practicing cloud financial management, or CFM, helps organizations realize business value and financial success as they optimize their cost and usage to scale on AWS. First, establish a cost optimization function. Create a team, such as a Cloud Business Office or Cloud Center of Excellence, that is responsible for establishing and maintaining cost awareness across your organization. The team requires participants from areas such as finance, technology, and business. Establish a partnership between finance and technology. Involve the finance and technology teams in cost and usage discussions at all stages of



your cloud journey. Teams can regularly meet and discuss topics such as organizational goals and targets, current cost and usage, and financial and accounting practices.

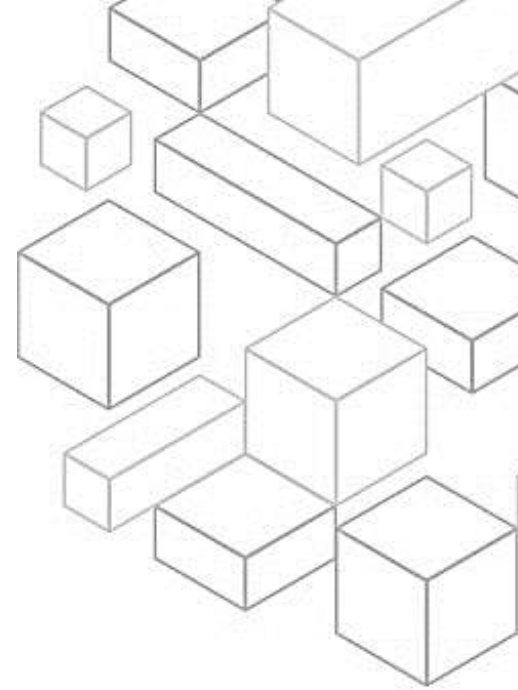
You can also establish cloud budgets and forecasts. Adjust existing organizational budgeting and forecasting processes to be compatible with the highly variable nature of cloud costs and usage. Processes must be dynamic, using trend-based or business-driver-based algorithms, or a combination. Implement cost awareness into new or existing organizational processes that impact usage, and consider existing processes for cost awareness. Implement cost awareness into employee training.

Next, report and notify on cost optimization. Configure AWS Budgets to provide notifications on cost and usage against targets. Hold regular meetings to analyze your workload cost efficiency and promote cost-aware culture. Monitor cost proactively for the workload by implement tooling and dashboards. Do not just look at costs and categories when you receive notifications. This helps to identify positive trends and promote them throughout your organization. Keep up-to-date with new service releases. Consult regularly with experts or AWS Partners to consider which services and features provide lower cost. Review AWS blogs and other information sources.

You can also create a cost-aware culture by implementing changes or programs across your organization. You can start small. Then, as your capabilities increase and your organization's use of the cloud grows, implement large and wide-ranging programs. Finally, quantify the business value of cost optimization, which helps you understand the entire set of benefits to your organization. Because cost optimization is a necessary investment, quantifying business value gives you the opportunity to explain the return on investment



to stakeholders. Quantifying business value can help you gain more buy-in from stakeholders on future cost optimization investments. It also provides a framework to measure the outcomes for your organization's cost-optimization activities.

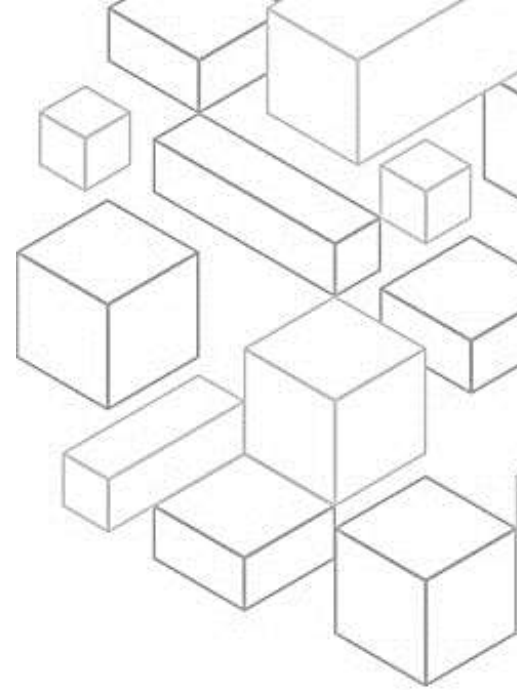


1.12 Expenditure and Usage Awareness

Expenditure and usage awareness is the next cost optimization best practice area. Understanding your organization's costs and drivers is critical to manage your cost and usage effectively and identify cost-reduction opportunities. Organizations typically operate multiple workloads run by multiple teams. These teams can be in different organization units, each with its own revenue stream. The capability to attribute resource costs to the workloads, individual organization, or product owners drives efficient usage behavior and helps reduce waste. Accurate cost and usage monitoring helps you understand how profitable organization units and products are. You can then make more informed decisions about where to allocate resources in your organization. Awareness of usage at all levels in the organization is key to driving change, as change in usage drives changes in cost.

1.13 Governance

Governance is used to establish policies and mechanisms to ensure that appropriate costs are incurred while objectives are achieved. By employing a checks-and-balances approach to governance, you can innovate without overspending. First, develop policies based on your organization requirements that define how resources will be managed. Policies should cover cost aspects



of resources and workloads, including creation, modification, and decommission over the resource lifetime.

Next, implement both cost and usage goals and targets for your workload. Goals provide direction to your organization on expected outcomes, and targets provide specific measurable outcomes to be achieved for your workloads.

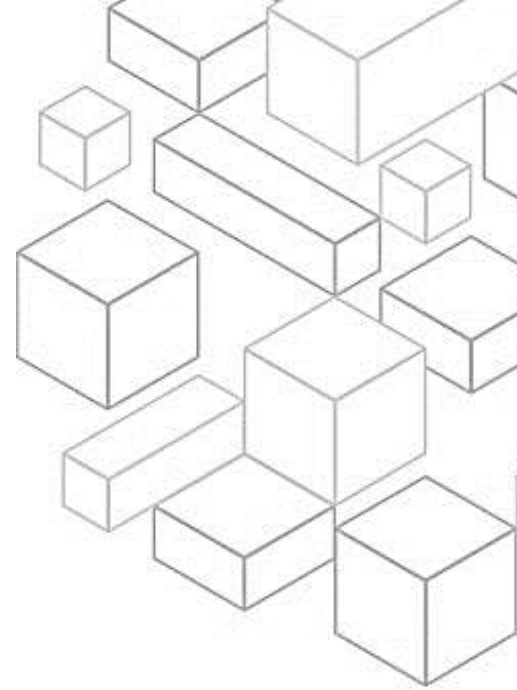
Implement a structure of accounts that maps to your organization. This assists in allocating and managing costs throughout your organization.

You can also implement groups and roles that align to your policies. This helps control who can create, modify, or decommission instances and resources in each group. For example, implement development, test, and production groups. This applies to both AWS services and third-party solutions. You can implement cost controls based on organization policies and defined groups and roles. These certify that costs are only incurred as defined by organization requirements. One example is control access to Regions or resource types with AWS Identity and Access Management, or IAM, policies.

Finally, track project lifecycle. You can track, measure, and audit the lifecycle of projects, teams, and environments to avoid using and paying for unnecessary resources.

1.14 Monitor cost and usage

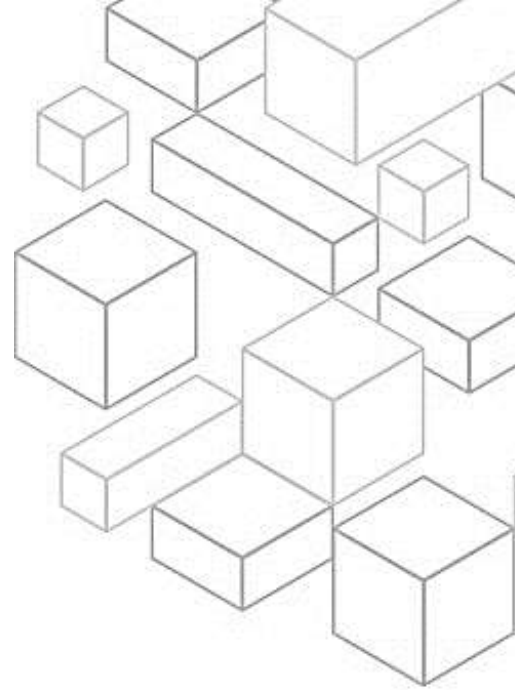
Monitor cost and usage to establish policies and procedures to monitor and appropriately allocate your costs. This helps you measure and improve the cost efficiency of this workload. The following are required areas for monitoring your cost and usage.



First, configure your information sources. To provide detailed cost and usage information, configure the AWS Cost and Usage Report and AWS Cost Explorer hourly granularity. Configure your workload to have log entries for every delivered business outcome. Next, add organization information to cost and usage. Define a tagging schema based on your organization, workload attributes, and cost allocation categories so that you can filter and search for resources or monitor cost and usage in cost management tools. Implement consistent tagging across all resources where possible by purpose, team, environment, or other criteria relevant to your business. You can also identify cost attribution categories. Identify organization categories that could be used to allocate cost inside your organization.

In addition, establish organization metrics. Identify the organization metrics that are required for this workload. Example metrics of a workload are customer reports produced or webpages served to customers. You can also configure cost management tools in line with your organization policies to manage and optimize cloud spend. This includes services, tools, and resources to organize and track cost and usage data, along with enhancing control through consolidated billing and access permission. You can also improve planning through budgeting and forecasts, receive notifications or alerts, and further lower cost with resources and pricing optimizations.

Finally, allocate the workload's costs by metrics or business outcomes to measure workload cost efficiency. Implement a process to analyze the AWS Cost and Usage Report with Amazon Athena, which can provide insight and charge-back capability.



1.15 Decommission resources

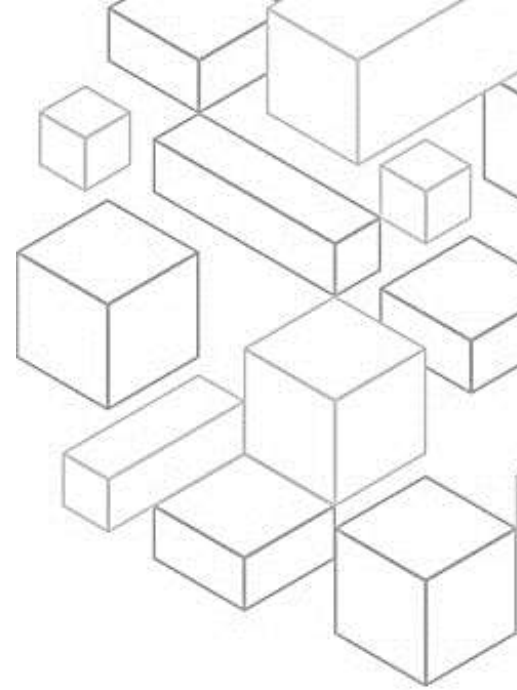
Decommission resources to implement change control and resource management from project inception to end of life. This helps ensure that you shut down or terminate unused resources to reduce waste. First, track resources over their lifetime. Define and implement a method to track resources and their associations with systems over their lifetime. You can use tagging to identify the workload or function of the resource.

You can also implement a decommissioning process to identify and decommission unused resources. Decommission resources invoked by events such as periodic audits or changes in usage. Decommissioning is typically performed periodically and can be manual or automated. You can also decommission resources automatically. Design your workload to gracefully handle resource termination as you identify and decommission noncritical resources, resources that are not required, or resources with low utilization.

Finally, enforce data retention policies. Define retention policies on supported resources to handle object deletion per your organizations' requirements. Identify and delete unnecessary or orphaned resources and objects that are no longer required.

1.16 Cost-Effective Resources

Cost-effective resources is the next cost optimization best practice area. Using the appropriate services, resources, and configurations for your workloads is key to cost savings.



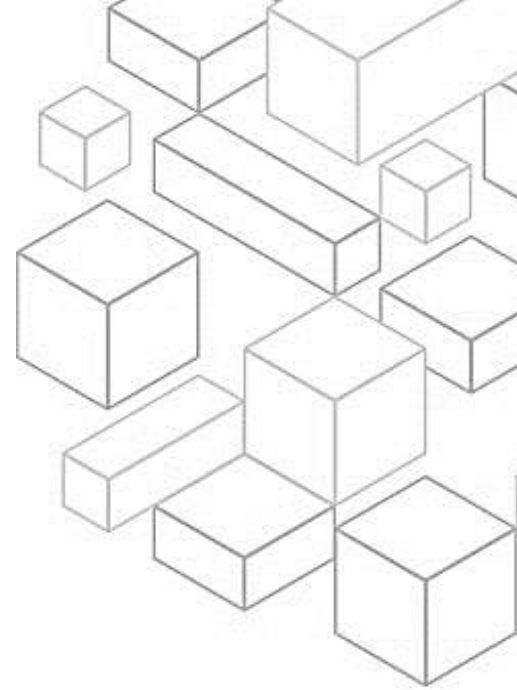
1.17 Evaluate cost when selecting services

Evaluate cost when selecting services. Amazon Elastic Compute Cloud, or Amazon EC2; Amazon Elastic Block Store, or Amazon EBS; and Amazon Simple Storage Service, or Amazon S3; are building-block AWS services. Managed services, such as Amazon Relational Database Service, or Amazon RDS, and Amazon DynamoDB, are higher-level, or application-level, AWS services. By selecting the appropriate building blocks and managed services, you can optimize a workload for cost. For example, using managed services, you can reduce or remove much of your administrative and operational overhead, freeing you to work on applications and business-related activities.

Consider the following best practices to help you evaluate cost when selecting the right services for your workload. First, identify organization requirements for cost. Work with team members to define the balance between cost optimization and other pillars, such as performance and reliability, for this workload. Then, analyze all components of the workload. Verify that every workload component is analyzed, regardless of current size or costs. The review effort should reflect the potential benefit, such as current and projected costs.

Next, perform a thorough analysis of each component. Look at overall cost to the organization of each component. Calculate the total cost of ownership by factoring in cost of operations and management, especially when using managed services by cloud provider. The review effort should reflect potential benefit.

You can also select software with cost-effective licensing. Open-source software eliminates software licensing costs, which can contribute significant costs to workloads. Where licensed software is required, avoid licenses bound

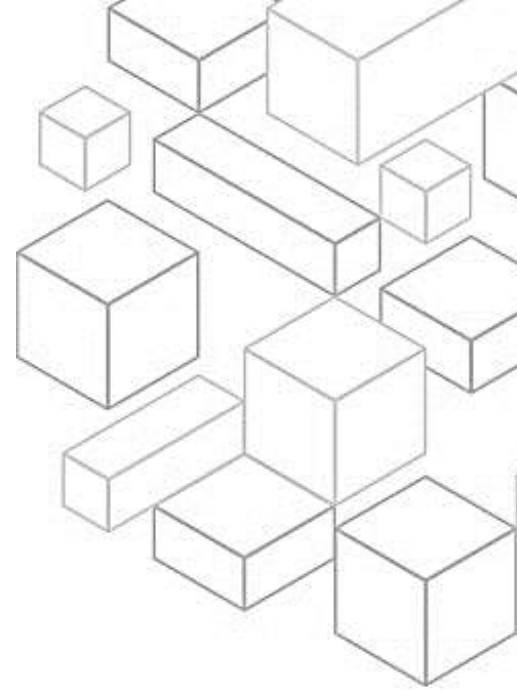


to arbitrary attributes such as CPUs and instead seek licenses that are bound to output or outcomes. The cost of these licenses scales more closely to the benefit they provide.

Select components of this workload to optimize cost in line with organization priorities. Factor in cost when selecting all components for your workload. This includes using application-level and managed services or serverless, containers, or event-driven architecture to help reduce overall cost. Minimize license costs by using open-source software, software that does not have license fees, or alternatives to reduce cost. Finally, perform cost analysis for workloads changing over time. Some services or features are more cost-effective at different usage levels. By analyzing each component over time and at projected usage, the workload can remain cost-effective over its lifetime.

1.18 Select correct resource type, size, and number

Select correct resource type, size, and number of resources to ensure that you have the appropriate configuration for the task at hand. You minimize waste by selecting the most cost-effective type, size, and number of resources. With AWS, there are a number of different approaches. First, perform cost modeling. Identify organization requirements, such as business needs and existing commitments. Then, perform cost modeling for overall costs of the workload and each component. Perform benchmark activities for the workload under different predicted loads, and then compare the costs. The modeling effort should reflect the potential benefit. For example, time spent is proportional to component cost.



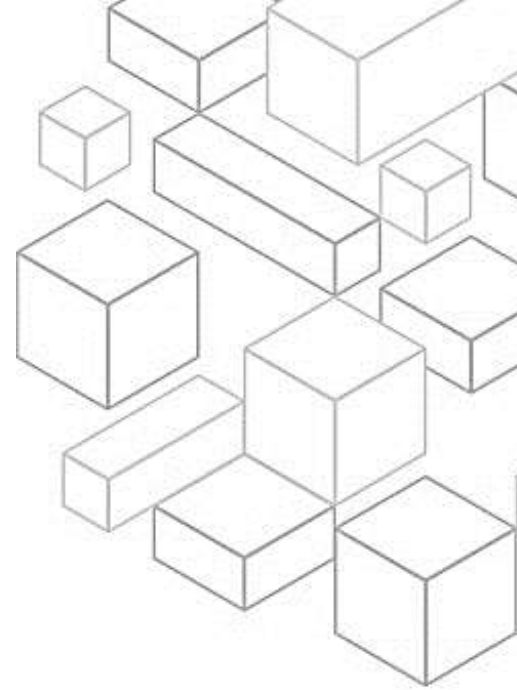
Second, select resource type, size, and number based on data about the workload and resource characteristics. Examples include compute, memory, throughput, or write intensive. This selection is typically made using a previous, on-premises version of the workload, documentation, or other sources of information about the workload.

Third, select resource type, size, and number automatically based on metrics. Use metrics from the currently running workload to select the right size and type to optimize for cost. Appropriately provision throughput, sizing, and storage for compute, storage, data, and networking services. This can be done with a feedback loop, such as automatic scaling or custom code in the workload.

1.19 Select pricing model

Select the pricing model that is most appropriate for your resources to minimize expense. Perform pricing model analysis by analyzing each component of the workload. Determine if the component and resources will be running for extended periods (for commitment discounts) or dynamic and short-running (for Spot or On-Demand). Perform an analysis on the workload using the recommendations in cost management tools. Then, apply business rules to those recommendations to achieve high returns.

Next, implement Regions based on cost. Resource pricing may vary by Region. Identify Regional cost differences, and only deploy in Regions with higher costs to meet latency and data sovereignty requirements. Factoring in Region cost helps you pay the lowest overall price for this workload.



Select third-party agreements with cost-efficient terms, which can help ensure that the cost of services scales with the benefits they provide. Select agreements and pricing that scale when they provide additional benefits to your organization. You can also implement pricing models for all components of this workload. Permanently running resources should use reserved capacity, such as Savings Plans or Reserved Instances. Short-term capacity is configured to use Spot Instances, or Spot Fleet. On-Demand Instances are only used for short-term workloads that cannot be interrupted and do not run long enough for reserved capacity, between 25 percent to 75 percent of the period, depending on the resource type.

Finally, perform pricing model analysis at the management account level. Check billing and cost management tools. Consider recommended discounts with commitments and reservations to perform regular analysis at the management account level.

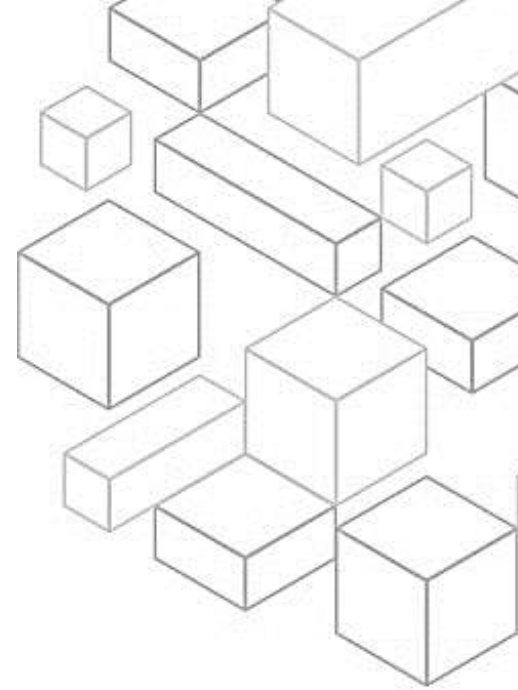
1.20 Plan for data transfer

Plan for data transfer to ensure that you plan and monitor data transfer charges so that you can make architectural decisions to minimize costs. A small yet effective architectural change can drastically reduce your operational costs over time. You can also gather organization requirements and perform data transfer modeling of the workload and each of its components. This identifies the lowest cost point for its current data transfer requirements.

Next, select components to optimize data transfer cost. All components are selected, and architecture is designed to reduce data transfer costs. This includes using components such as wide area network (WAN) optimization and



multiple Availability Zone configurations. Finally, select services to reduce data transfer costs. For example, use a content delivery network such as Amazon CloudFront to deliver content to end users, cache layers using Amazon ElastiCache, or try AWS Direct Connect instead of virtual private network (VPN) to connect to AWS.

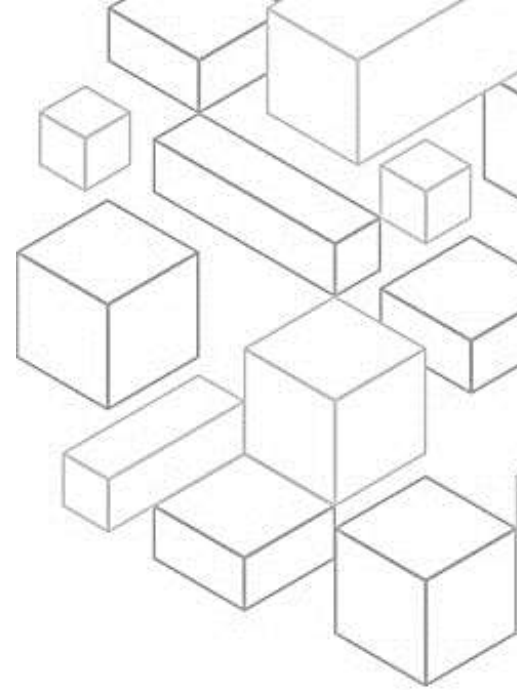


1.21 Manage Demand and Supply Resources

Manage demand and supply resources is the next cost optimization best practice area. When you move to the cloud, you pay only for what you need. You can supply resources to match the workload demand at the time they are needed - eliminating the need for costly and wasteful over-provisioning. You can also modify the demand using a throttle, buffer, or queue to smooth the demand and serve it with less resources. The economic benefits of just-in-time supply should be balanced against the need to provision to account for resource failures, high availability, and provision time. Depending on whether your demand is fixed or variable, plan to create metrics and automation that will ensure that management of your environment is minimal - even as you scale. When modifying the demand, you must know the acceptable and maximum delay that the workload can allow.

1.22 Manage demand and supply resources

Manage demand and supply resources. For a workload that has balanced spend and performance, ensure that everything you pay for is used and avoid significantly under-utilizing instances. A skewed utilization metric in either direction can adversely impact your organization. This could impact



operational costs in the form of degraded performance due to over-utilization or wasted AWS expenditures due to over-provisioning.

The following best practices describe how to use these approaches. First, perform an analysis on the workload demand. Analyze the demand of the workload over time. Verify that the analysis covers seasonal trends and accurately represents operating conditions over the full workload lifetime. Analysis effort should reflect the potential benefit, such as spending time proportional to the workload cost.

Second, implement a buffer or throttle to manage demand. Buffering and throttling modify the demand on your workload, smoothing out any peaks. Implement throttling when your clients perform retries. Implement buffering to store the request, and defer processing until a later time. Verify that your throttles and buffers are designed so that clients receive a response in the required time.

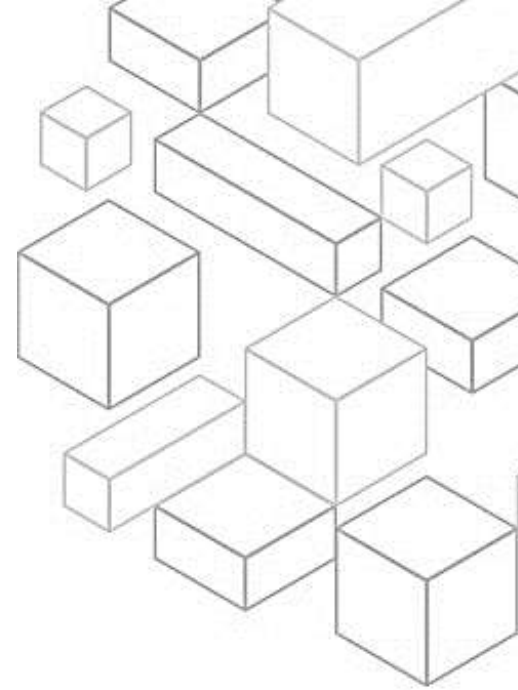
Finally, supply resources dynamically. Resources are provisioned in a planned manner. This can be demand-based, such as through automatic scaling, or time-based, where demand is predictable and resources are provided based on time. These methods result in the least amount of over-provisioning or under-provisioning.

1.23 Optimize Over Time

Optimize over time is the next cost optimization best practice area. With AWS, you optimize over time by reviewing new services and implementing them in your workload. As AWS releases new services and features, it is a best practice



to review your existing architectural decisions to help ensure that they remain cost-effective. As your requirements change, take action to decommission resources, components, and workloads that you no longer require.



1.24 Optimize over time

Optimize over time. As AWS releases new services and features, it's a best practice to review your existing architectural decisions to ensure that they continue to be the most cost-effective. First, develop a workload review process that defines the criteria and process for review. The review effort should reflect the potential benefit. For example, core workloads or workloads with a value of over 10 percent of the bill should be reviewed quarterly or every 6 months. Workloads below 10 percent can be reviewed annually.

Second, review and analyze this workload frequently. Existing workloads are regularly reviewed based on each defined process. This helps determine whether new services can be adopted, existing services can be replaced, or workloads can be re-architected.

Finally, perform automations for operations. Evaluate the cost of effort for operations in the cloud. Quantify reduction in time and effort for administrator tasks, deployment, and other operations using automation. Evaluate the required time and cost for the effort of operations, and automate admin tasks to reduce the human effort where possible.



1.27 Summary

In this module, you learned about the cost optimization pillar. This started with an overview and included in depth discussion on the value proposition, design principles, and best practices of the cost optimization pillar.

1.28 Thank you

Thank you for your participation!

